

REDROB · INTELLIGENT CANDIDATE DISCOVERY & RANKING

Reading between the lines

A candidate ranker that hires the way a great recruiter does —
by understanding what the role means, not by matching keywords.

100K

candidates

Top 100

trusted shortlist

Hybrid

dense + reasoning

36 s

CPU, no network

THE PROBLEM

Keyword filters can't see what actually matters

1

Recruiters scan hundreds of profiles

and still miss the right person — not for lack of talent, but because filters match words, not fit.

2

The JD says one thing and means another

“5–9 years” is a guide, not a gate. “AI skills” without the right title is a red flag, not a green one.

3

The best fit may never say the buzzword

Someone who built “the ranking systems that decide what users see” is a fit even if they never write “RAG”.

One narrow role, a 100,000-profile haystack

What the JD really wants

- › Production embeddings retrieval + vector search
- › Shipped a ranking / recsys / search system at scale
- › At a product company (not services-only)
- › Evaluation rigor: NDCG, MRR, MAP, A/B testing
- › 5–9 yrs, pragmatic shipper > research-only
- › In or willing to relocate to Noida / Pune, India

100,000

candidate profiles

~1,000

genuine ML/AI engineers

~5,500

keyword-stuffer traps

~80

impossible honeypots

Scored on a hidden, graded ground truth

$$0.50 \cdot \text{NDCG@10} + 0.30 \cdot \text{NDCG@50} + 0.15 \cdot \text{MAP} + 0.05 \cdot \text{P@10}$$

Top-10 quality is half the score — precision at the very top is everything.

Same keywords. Opposite verdicts.

✗ KEYWORD-STUFFER

"Project Manager · AI enthusiast · Building with LLMs"

Title & career: PM / Marketing / Sales

Skills list sprayed with RAG, Pinecone, FAISS

Summary: "took online courses on RAG..."

→ buzzwords are cheap; the work isn't there

our score: 0.04 · buried

✓ PLAIN-LANGUAGE TIER-5

"...the ranking and retrieval systems that decide what to show"

Title & career: Senior AI Eng @ product co.

Skills in plain words: Ranking Systems, BM25

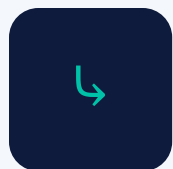
Summary maps 1:1 onto the JD's mandate

→ never says "RAG", but clearly built it

our score: 0.80 · top of the list

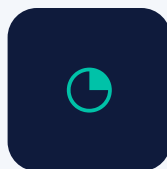
THE INSIGHT THAT DRIVES EVERYTHING

Buzzwords are noise. Evidence and availability are signal.



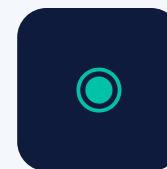
Read the career, not the skills list

Credit for an AI skill is GATED by title/career credibility. A Marketing Manager's "RAG" earns almost nothing; an engineer's earns full weight.



A quiet tell in the data

"Python" appears as a listed skill on only ~1.4% of profiles — the genuine engineers. Stuffers reach for the flashy buzzwords instead.



Hireable ≠ qualified

A perfect profile that hasn't logged in for 6 months with a 5% response rate isn't available. Behaviour modifies fit — it never manufactures it.

ARCHITECTURE

A hybrid: dense recall, structured precision



Why hybrid?

Embeddings give recall over fits described in plain language. The structured + integrity layers give precision and keep stuffers and honeypots out of the top. Neither wins alone — which is exactly the system this role is hired to build.

One transparent, auditable formula

$$\text{final} = \text{fit} \times \text{behavioural} \times \text{disqualifier} \times \text{honeypot} \times \text{stuffer}$$

fit = weighted blend



Multipliers

Behavioural availability	× 0.62 – 1.08
CV / speech, no IR-NLP	× 0.55
Research-only, no production	× 0.60
Keyword-stuffer	× 0.25 – 0.40
Honeypot (impossible profile)	× 0.03

Every trap has a dedicated guard

TRAP	HOW IT HIDES	GUARD
 Keyword-stuffer	Non-eng title + buzzword skills	Gate skill-credit on title credibility; anti-query dampener
 Honeypot	Tenure > dates allow; “expert”, 0 months used	Date-arithmetic & mastery-without-time checks → ×0.03
 CV / speech specialist	YOLO, OpenCV, ASR — no retrieval	Penalise CV-heavy profiles lacking IR/NLP evidence
 Research-only	Papers, labs, no deployment	Require production signal or down-weight
 Title-chaser	Job-hops every <18 months for titles	Tenure-stability term in the career-arc score

Every pick comes with honest, grounded reasoning

GENERATED REASONING (rank 65)

“AI Engineer with 5 yrs (Salesforce); direct ranking systems, recommendation systems experience (Haystack, LoRA, Weaviate); Pune-based. Concern: long notice period (120d).”

Facts pulled only from the profile — no LLM, no invented skills.



Specific facts: title, years, real employers



Connects to JD: names the retrieval/ranking work



Honest: flags the 120-day notice concern



No hallucination: skills are from the profile



Varied: composed per candidate, not templated



Tone matches rank: confident high, hedged low

RESULTS ON THE RELEASED POOL

A shortlist a recruiter can trust

100%

engineering titles in the top 100

86 / 100

inside the 5–9 yr band (mean 6.5)

0

honeypots & 0 stuffers in top 100

Noida + Pune

the #1 and #2 locations

Disqualification-proof: the spec fails any submission with >10% honeypots in the top 100. Ours has zero.

BUILT FOR PRODUCTION, NOT JUST A BENCHMARK

Fast, reproducible, no API in the loop

1

Offline · once

Embed 100K profiles with a small CPU transformer; build the FAISS index. ~20 min, fully documented.

2

Ranking step · every run

Pure numpy over the pre-computed index. 36 seconds for 100K on a 12-core CPU — far inside the 5-min budget.

3

No network, no GPU, no LLM calls

The ranking code makes zero API calls — the same constraint a 200K-pool production system must meet.

ONE COMMAND

```
python rank.py
  --candidates candidates.jsonl
  --out submission.csv
```

Runtime	≤ 5 min	✓	36 s
Memory	≤ 16 GB	✓	
Compute	CPU only	✓	
Network	off	✓	

WHY IT WINS

Understanding, not keyword counting

- Hybrid dense + structured ranking — recall and precision, not one at the cost of the other.
- Engineered against every trap the dataset hides; zero honeypots and zero stuffers in the top 100.
- Optimised where it counts: top-10 quality is half the score, so the top is all in-band, on-domain fits.
- Fast, reproducible, and fully auditable — every score and every reason can be explained line by line.

Make hiring smarter — rank candidates the way a great recruiter would.